Enrolment Number										

Total No. of printed pages = 04

Winter, 2026

B. Tech 4th Semester End Semester Examination

Geotechnical Engineering– Theory

Course Code: BCE23403T

Full Marks – 50 **Time – 2 hours**

The figure in the margin indicates full marks for the questions.

SECTION 1

Answer all the questions **Marks: 10 x 1 =10**

1. (a) Soil deposited in lakes is known as:

(i) Marine soil (ii) Lacustrine soil

(iii) Aeolian soil (iv) Glacial soil

(b) The ratio of volume of voids to volume of solids is called:

(i) Porosity (ii) Water content

(iii) Void ratio (iv) Specific gravity

(c) For a saturated soil:

(i) S = 0% (ii) S = 50%

(iii) S = 100% (iv) S > 100%

(d) Quick sand condition occurs when:

(i) Hydraulic gradient becomes zero (ii) Effective stress becomes zero

(iii) Total stress becomes zero (iv) Water content becomes zero

(e) Which soil generally possesses the highest permeability?

(i) Clay (ii) Silt

(iii) Gravel (iv) Peat

(f) The test best suited for measuring undrained strength of soft clay is

(i) Direct shear test (ii) Triaxial test

(iii) Vane shear test (iv) Plate load test

(g) In a flow net, flow lines and equipotential lines intersect at:

(i) 30° (ii) 45°

(iii) 60° (iv) 90°

(h) The relationship between unconfined compressive strength and cohesion is:

(i) qu = c (ii) qu = 4c

(iii) qu = 2c (iv) qu = 0.5c

(i) The optimum moisture content corresponds to:

(i) Minimum dry density (ii) Maximum dry density

(iii) Zero air voids (iv) Maximum saturation

(j) Settlement due to expulsion of pore water is called:

(i) Immediate settlement (ii) Consolidation settlement

(iii) Elastic settlement (iv) Differential settlement

SECTION 2

Answer any one question from this section **Marks: 10**

2(a) Explain the various methods of soil exploration. A sand deposit is characterized by a maximum void ratio of 0.82 and a minimum void ratio of 0.46. If the in-situ void ratio of the deposit is 0.58, calculate the relative density of the sand and if the specific gravity of soil solids is 2.65, calculate the dry unit weight of the deposit. 4+6 = 10

2(b) Define disturbed and undisturbed samples. An undisturbed cylindrical soil specimen has a volume of 150 cm^3 and a wet mass of 270 g. After oven drying, its mass is reduced to 225 g. If the specific gravity of soil solids is 2.70, determine the water content, void ratio, and degree of saturation of the soil specimen. **4+6 = 10**

SECTION 3

Answer any one question from this section **Marks: 10**

3 (a)Explain the significance of flow nets in seepage analysis. A cofferdam is founded on a homogeneous isotropic soil deposit. The difference in water level between the upstream and downstream sides is 12 m. A flow net constructed for the cofferdam consists of 6 flow channels and 18 potential drops. If the coefficient of permeability of the soil is 3×10^{-5} cm/s, determine the seepage discharge per metre length of the cofferdam. **5+5=10**

(b) Discuss the factors affecting permeability of soils. A layered soil profile consists of:

Layer	Thickness (m)	Permeability (cm/s)
1	5	2×10^{-4}
2	8	6×10^{-4}
3	7	10×10^{-4}

Determine: **(5+5=10)**

a) Equivalent horizontal permeability

b) Equivalent vertical permeability

SECTION 4

Answer any one question from this section **Marks: 10**

4 (a) Define normally consolidated and overconsolidated clay. A clay layer has a thickness of 6 m and is subjected to an increase in pressure of 80 kN/m². The layer is drained on both sides and attains 50% consolidation in a year. Determine the coefficient of consolidation taking $T_v = 0.197$. If the coefficient of permeability of the clay is 0.03 m/year, determine:

i) The final consolidation settlement of the clay layer.

ii) The settlement that occurs after a year. **(3+3+2+2=10)**

(b) Define Zero Air Void (ZAV) line. Using the compaction test data provided, calculate the dry density values and draw the compaction curve. Determine the optimum moisture content and the corresponding maximum dry density of the soil. **(3+3+3+1=10)**

Water content (%)	9	11	13	15	17
Wet density (g/cc)	1.84	1.95	2.02	1.98	1.92

SECTION 5

Answer any one question from this section **Marks: 10**

5 (a) Define pressure bulb. A circular footing of 8 m diameter transmits a uniformly distributed pressure of 50 kN/m² to the ground surface. Using Boussinesq's and Westergaard's analysis, determine the vertical stress at a depth of 5 m below the centre of the footing. **(2+4+4=10)**

(b) What are the advantages of the triaxial compression test over the direct shear test? A standard specimen of cohesionless sand was tested in a drained triaxial compression test. The specimen failed at a deviator stress of 540 kN/m² under a cell pressure of 120 kN/m².